EVERFRESH · GROW CHAMBER V2

The Vitrine

A LIVING SHOWPIECE CABINET



A warm-wood-frame, all-glass grow cabinet on casters — built to grow one Cojoba angustifolia hard, and to be beautiful doing it.

~24 × 18 × 52 IN • ~$200 + LED • ~10-DAY BUILD

THE IDEA

A cabinet the tree lives in



v1 was a wire rack wrapped in taped plastic — leaky, fogged, awkward to open, and no one's idea of a centerpiece. **v2 flips the goal:** a glass-on-all-four-sides conservatory cabinet that reads as furniture. More room for the leader that's already arching for the light, a full front door for easy daily access, and the glowing tree as the whole point of the room.

INTERIOR	VOLUME	CABINET BUDGET	TIMELINE
24×18×52 in	**~14.6** cu ft	**~$200** +LED	**~10** days

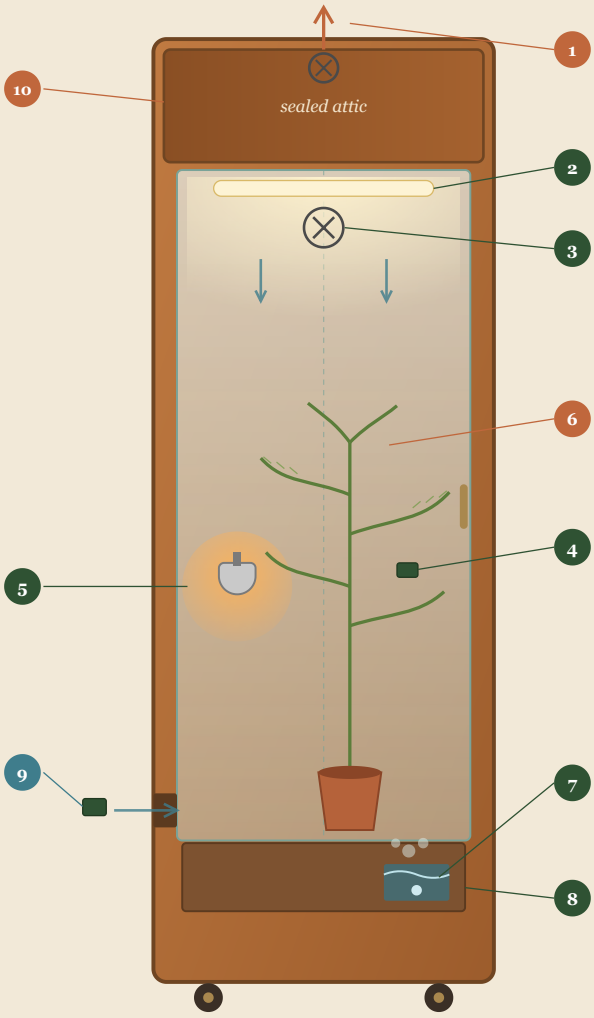
PRIORITIES, IN ORDER

1 More room · 2 Easy access + beautiful · 3 Leak-sealing · 4 Insulation (out by design)

Glass on all sides means insulation can't come along – and it's your last priority anyway. The trade: heat & light spill to the room (the glow is the feature). Keep the west window shaded for the 4–8 PM solar spike, still the highest-leverage thermal fix.

ANATOMY

How it all fits



1 Active top exhaust
PC fan in a port cut to the fan body (foam collar, no short-circuit) — the peak-shaver during the afternoon solar spike.

2 Full-spectrum white LED
Sleek dimmable panel under the top; driver hidden in the attic. Warm daylight glow, not blurple.

3 Circulation fan
Blows down to mix the gradient; gentle continuous trickle between cycles.

4 Canopy sensor
SHT31 at leaf height — the control point every decision uses.

5 Reptile heat bulb
Mid-level above the reservoir, radiating up. Your proven heat source.

6 The tree
Cojoba angustifolia — room to reach ~3 ft before the next rebuild.

7 Fogger + reservoir
Humidity source; refill through the front door.

8 Waterproof base pan
Sealed, sloped to a corner — catches fog-spit + condensation.

9 Passive intake + outdoor sensor
Low screened port; second SHT31 reads the air the exhaust pulls in.

10 Sealed attic
All electronics — exhaust fan, LED driver, Photon — out of the humid zone.

What carries over & what changes

Keep from v1	Change for v2
Fogger + reservoir (humidity)	Drop heat mat & intentional floor pool
Reptile heat bulb, mid-level, radiating up	Grow light → sleek dimmable white LED
Circ fan up top, blowing down	Enclosure → wood + glass vitrine
Passive low intake + active top exhaust	Access: taped flap → full swing door
2× SHT31 (canopy + intake) · Photon	Hidden waterproof base pan for runoff

No firmware changes needed – sensor roles, actuators, and airflow direction are all unchanged.

Lighting — white, not blurple

- **ViparSpectra P1000** ~$100 – recommended
- Mars Hydro TS600 ~$70 – value
- Barrina bars ~$35 – sleekest

Dimmable (hormesis tuning). Driver in the attic. Target ~200–400 PPFD.

Seal the leaks

- EPDM **bulb gasket** around the door
- **Compression / roller latch** pulls it tight
- Panels bedded in 100% silicone
- All wires via **sealed glands** in the attic

Aesthetic — warm & tropical

- **Frame:** Douglas fir (honey) or cedar (rot-resistant, reddish); sapele if you splurge
- **Finish:** teak/Danish oil → spar urethane – warmth + waterproofs the wood in lieu of a liner
- **Hardware:** brass hinge, handle & latch – conservatory feel

AIRFLOW – THE NEGATIVE-PRESSURE BOX

A **Passive intake, low** – a screened port draws room air in at the base. → B **Circ fan pushes down** – mixes the gradient; heat bulb warms the rising column. → C **Active exhaust, top** – pulls stale/hot air out; the afternoon peak-shaver.

Sealed tight, this finally lets the CO_2 vent schedule come back on – the airflow goal v1 never quite reached.

THE BUILD

Budget & plan



Item	Est.
Acrylic, 1/8″ (3/16″ door), ~30 sq ft cut to size — TAP Plastics	$90
Frame wood — Douglas fir / cedar (2×2 posts, 1×2 rails)	$40
Casters ×4 (2 locking)	$18
Piano hinge + brass handle + latch	$22
EPDM bulb weatherstrip	$10
Silicone ×2 + glue + fasteners	$15
Teak oil + spar urethane	$20
Cabinet total (LED separate)	**~$215**

BUILD SEQUENCE

1. Order/cut acrylic at TAP (lead time — first)
2. Mill + dry-fit frame; cut joinery + panel rabbets
3. Finish wood (oil + spar) *before* glazing
4. Build base pan + attic; cut exhaust/intake ports
5. Glaze panels in silicone; hang door + gasket + latch
6. Fogger leak-check by eye before wiring
7. Mount LED, fans, heat bulb, fogger, sensors
8. Power up, seal-test, tune

Three calls to lock it

1. **Footprint** — 24×18 (my pick) or 20×20 square?
2. **LED** — ViparSpectra P1000 (recommended), TS600, or Barrina bars?
3. **Frame wood** — Douglas fir (cheap, honey), cedar (rot-proof), or sapele (splurge)?